

Exercises Chapter 2

1. Consider the following linear regression

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \epsilon_i,$$

where $X_{i1} = 1$ for all $i = 1, \dots, n$.

- (a) Show that Assumption 2 (exogeneity) is violated if some of the regressors are correlated (i.e. $\text{Cov}(X_{ik}, \epsilon_i) \neq 0$ for at least one $k = 0, \dots, p$) with the error term.

2. (a) Prove that

$$\bar{X} = \arg \min_{a \in \mathbb{R}} \sum_{i=1}^n (X_i - a)^2$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean of the iid random sample $\{X_1, \dots, X_n\}$.

- (b) Prove that

$$\sum_{i=1}^n (X_i - \bar{X})^2 \leq \sum_{i=1}^n (X_i - \mu)^2,$$

where $\mu = E(X_i)$ is the population mean of $X_1, \dots, X_n$.

3. Suppose we have a data set with dependent variable

$$Y = \text{“starting salary after graduation”}$$

and four predictors

- $X_1 = \text{GPA}$
- $X_2 = \text{IQ}$
- $X_3 = \text{Gender}$ (1 for Female and 0 for Male)
- $X_4 = X_1 X_2$ Interaction between GPA and IQ
- $X_5 = X_1 X_3$ Interaction between GPA and Gender

Suppose we use least squares to fit the model and get the following estimates:

$$\hat{\beta}_0 = 12, \hat{\beta}_1 = 24, \hat{\beta}_2 = 0.04, \hat{\beta}_3 = 20, \hat{\beta}_4 = 0.04, \text{ and } \hat{\beta}_5 = -8.$$

Thus we have:

$$\hat{E}[Y|X] = \hat{f}(X) = 12 + 24 \overbrace{\text{GPA}}^{X_1} + 0.04 \overbrace{\text{IQ}}^{X_2} + 20 \overbrace{\text{Gender}}^{X_3} + 0.04 \overbrace{\text{GPA} \cdot \text{IQ}}^{X_4 = X_1 \cdot X_2} - 8 \overbrace{\text{GPA} \cdot \text{Gender}}^{X_5 = X_1 \cdot X_3}$$

Show that, for a fixed value of IQ and GPA, males earn more on average than females provided GPA is greater than 2.50

4. (a) Consider the linear regression model $Y = X\beta + \epsilon$. Let

$$Y = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \quad \text{and let} \quad M_X = \begin{pmatrix} 0.5 & 0.0 & -0.5 \\ 0.0 & 0.0 & 0.0 \\ -0.5 & 0.0 & 0.5 \end{pmatrix}$$

What is the value of the residual sum of squares $\sum_{i=1}^3 e_i^2$?